



MATERIAL SAFETY DATA SHEET

1. Chemical Product Identification

Active Ingredient (%): S-Metolachlor 96% W/V EC

Chemical Name:

Acetamide, 2-chloro-N-(2-ethyl-6-methylphenyl)-N-(2-methoxy-1-methylethyl)-, (S)

Chemical Class: Chloroacetanilide Herbicide

Manufacturer: SHANDONG BINNONG TECHNOLOGY CO., LTD.

NO.518, Yongxin Road, Binbei Town, Binzhou, Shandong, China

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2. Composition / Information on Ingredients

Ingredient	CAS NO	Percent (w/v)
S-metolachlor	51218-45-2	96.0%
Inert ingredient	/	Up to 100%

Relative density: 1.10 g/cm³ at 20 -25°C.

3. Hazards Identification

Symptoms of Acute Exposure

Causes eye, skin and respiratory passage irritation. Allergic skin reactions are possible. Exposure to high vapor levels may cause headache, dizziness, numbness, nausea, incoordination, or other central nervous system effects.

Hazardous Decomposition Products

Can decompose at high temperatures forming toxic gases.

Unusual Fire, Explosion and Reactivity Hazards

During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

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REG.DEP.: 86-543-3363985



4. First Aid Measures

Have the product container, label or Material Safety Data Sheet with you when calling a poison control center or doctor, or going for treatment

Ingestion: If swallowed: Call a poison control center or doctor immediately for treatment advice. Do not give any liquid to the person. Do not induce vomiting unless told to do so by a poison control center or doctor. Do not give anything by mouth to an unconscious person

Eye Contact: If in eyes: Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after 5 minutes, then continue rinsing eye. Call a poison control center or doctor for treatment advice.

Skin Contact: If on skin or clothing: Take off contaminated clothing. Rinse skin immediately with plenty of water for 15- 20 minutes. Call a poison control center or doctor for treatment advice.

Inhalation: If inhaled: Move person to fresh air. If person is not breathing, call 911 or an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a poison control center or doctor for further treatment advice

Notes to Physician

There is no specific antidote if this product is ingested.

Treat symptomatically.

Persons suffering a temporary allergic reaction may respond to treatment with antihistamines or steroid creams and/or systemic steroids.

Contains petroleum distillate - vomiting may cause aspiration pneumonia.

5. Fire-Fighting Measures

Fire and Explosion

Flash Point (Test Method): > 93°C

Flammable Limits (% in Air): Lower: % Not Applicable Upper: % Not Applicable

Autoignition Temperature: Not Available

Flammability: Not Applicable

Autoignition temperature: Not determined.

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Hazardous combustion products: Oxides of carbon, nitrogen and sulphur.

Extinguishing media: Carbon dioxide, dry chemical, foam and water fog.

Fire-fighting instructions: Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion. Prevent use of contaminated buildings, area and equipment until decontaminated.

Unusual Fire, Explosion and Reactivity Hazards

During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion

In Case of Fire:

Use dry chemical, foam or CO₂ extinguishing media. Wear full protective clothing and self-contained breathing apparatus.

Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion.

Prevent use of contaminated buildings, area, and equipment until decontaminated. Water runoff can cause environmental damage. If water is used to fight fire, dike and collect runoff.

6. Accidental Release Measures

In Case of Spill or Leak

Control the spill at its source. Contain the spill to prevent from spreading or contaminating soil or from entering sewage and drainage systems or any body of water. Clean up spills immediately, observing precautions outlined in Section 8. Cover entire spill with absorbing material and place into compatible disposal container. Scrub area with hard water detergent (e.g. commercial products such as Tide, Joy, Spic and Span). Pick up wash liquid with additional absorbent and place into compatible disposal container. Once all material is cleaned up and placed in a disposal container, seal container and arrange for disposition.

7. Handling And Storage

Store the material in a well-ventilated, secure area out of reach of children and domestic animals.

Do not store food, beverages or tobacco products in the storage area. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the

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material. Wash thoroughly with soap and water after handling.

8. Exposure Controls/Personal Protection

Ingestion: Prevent eating, drinking, tobacco usage and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling

Eye Contact: Where eye contact is likely, use chemical splash goggles. Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

Skin Contact: Where contact is likely, wear chemical-resistant (such as nitrile or butyl) gloves, coveralls, socks and chemical-resistant footwear. For overhead exposure, wear chemical-resistant headgear.

Inhalation: A respirator is not normally required when handling this substance. Use effective engineering controls to comply with occupational exposure limits.

In case of emergency spills, use a NIOSH approved respirator with any R, P or HE filter.

9. Physical and Chemical Properties

Appearance: Yellow brown liquid

Odor: Faint odor

Flash Point: 85°C at 10-1.5 kPa

Melting Point: Not Applicable

Boiling Point: Not Available

Specific Gravity/Density: 1.09-1.13 g/cm³ (20°C)

pH: 5-9

Persistence foam: Max. 60 ml

Emulsion stability: The emulsion is stable, there are no separate layers.

10. Stability and Reactivity

Stability: Stable under normal use and storage conditions

Hazardous Polymerization: Will not occur.

Conditions to Avoid: None known.

Materials to Avoid: None known

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Hazardous Decomposition Products: Can decompose at high temperatures forming toxic gases.

11. Toxicological Information

Acute Toxicity/Irritation Studies (Finished Product)

Ingestion: Slightly Toxic

Oral (LD₅₀ Rat): > 2000 mg/kg body weigh

Dermal: Slightly Toxic

Dermal (LD₅₀ Rabbit) : > 2000 mg/kg body weight

Inhalation: Practically Non-Toxic

Inhalation (LC₅₀ Rat): > 3.19 mg/l air - 4 hours

Eye Contact: No eye to Irritant (Rabbit)

Skin Contact: No skin to Irritant (Rabbit)

Skin Sensitization: moderate-sensitizer to skin (Guinea Pig)

Reproductive/Developmental Effects

S-Metolachlor: None observed.

Chronic/Carcinogenicity

12. Ecological and Ecotoxicological Information

Summary of Effects

S-Metolachlor:

Slightly to moderately toxic to fish.

Slightly toxic to invertebrates.

Practically non-toxic to birds and bees.

Eco-Acute Toxicity

S-Metolachlor: Bees LC₅₀/EC₅₀ > 200 ug/bee

Invertebrates (Water Flea) LC₅₀/EC₅₀ 26 ppm

Fish (Trout) LC₅₀/EC₅₀ 12 ppm

Fish (Bluegill) LC₅₀/EC₅₀ 3.16 ppm

Birds (8-day dietary - Bobwhite Quail) LC₅₀/EC₅₀ > 5,620 ppm

Birds (8-day dietary - Mallard Duck) LC₅₀/EC₅₀ > 5,620 ppm

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Eco-Chronic Toxicity

S-Metolachlor: Not Available

Environmental Fate

S-Metolachlor:

The information presented here is for the active ingredient, S-metolachlor. Low bioaccumulation potential. Not persistent in soil. Stable in water. Sinks in water (after 24 h).

13. Disposal Considerations

Disposal:

Do not reuse product containers. Dispose of product containers, waste containers, and residues according to local, state, and federal health and environmental regulations.

Characteristic Waste:

Under certain circumstances, discarded product may exhibit TCLP hazardous characteristics. A hazardous waste determination should be done on a case by case basis

Listed Waste:

Not Applicable

14. Transport Information

Hazard class : 9

UN NO. : 3082

15. Regulatory Information

N – Dangerous for environment

Xi – Irritating

R – 43

R – 50/53 – Very toxic to aquatic organisms, may cause long term adverse effects in the aquatic environment.

S – 2 – Keep out of the reach of children

S – 24 - Avoid contact with skin

S – 37 - Wear protective gloves

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16. Other Information

All information and instructions provided in this Material Safety Data Sheet (MSDS) are based on the current state of scientific and technical knowledge at the date indicated on the present MSDS and are presented in good faith and believed to be correct. This information applies to the product as such.

In case of new formulations or mixes, it is necessary to ascertain that a new danger will not appear. It is the responsibility of persons on receipt of this MSDS to ensure that the information contained herein is properly read and understood by all people who may use, handle, dispose or in any way come in contact with the product. If the recipient subsequently produce formulations containing this product, it is the recipients sole responsibility to ensure the transfer of all relevant information from this MSDS to their own MSDS.

In case of emergency please contact;
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